

Supplementary Material

EATTA + CAS: Abrupt Stream Evaluation Across Three Seeds

Lahore University of Management Sciences — Undergraduate Research Project

This document accompanies the main paper as supplementary material. It presents the complete per-seed evaluation results for the abrupt stream which includes per-batch accuracy curves, BN anchor novelty diagnostics, OATTA gate behaviour, per-corruption breakdowns, and CAS selection analysis.

All results are reported for three independent random seeds (42, 123, 456) to assess statistical reliability. A consolidated summary and full list of research-positive findings are provided at the end.

The Abrupt Stream

The abrupt stream is the primary evaluation stream in our experimental setup. It is designed to maximize the number of clean, instantaneous corruption-type transitions available for the BN anchor to detect, making it the most direct test of the core CAS mechanism. Six corruption types are presented sequentially, each at a uniform severity of 3, with 5000 images per block and a batch size of 64, giving 78 batches per block and 469 total batches across the full stream.

The corruption sequence is:

Corruption	Severity	Images	Batches
fog	3	5000	78
gaussian_noise	3	5000	78
motion_blur	3	5000	78
frost	3	5000	79
contrast	3	5000	78
jpeg_compression	3	5000	78
Total		30,000	469

Using a single severity level throughout ensures that cross-block differences in BN statistics arise purely from the corruption *type*, not from a confound between corruption type and severity. This maximizes the BN anchor’s signal-to-noise ratio and provides the cleanest possible test of whether the anchor can discriminate environmental transitions. All five transitions are abrupt and instantaneous: the final batch of one corruption block is immediately followed by the first batch of the next, with no mixing or gradual crossover.

All images are processed under a predict-then-learn protocol. Accuracy is recorded before adaptation occurs on each batch, matching the realistic online inference setting. We evaluate Online Mean Accuracy – the mean per-batch accuracy across all 469 batches.

Four methods are compared simultaneously: **Source** (unadapted ResNet-50), **EATTA** (base active test-time adaptation), **CAS pre-OATTA** (EATTA with BN anchor and context-aware selection), and **CAS+OATTA** (the full system with LLR-gated Bayesian filter).

The stream is evaluated under three independent random seeds (42, 123, 456), each shuffling the within-block sample ordering differently while preserving the corruption block structure.

Seed 42 (stream_abrupt_seed42)

Overall Accuracy Summary

Method	Online Mean Acc.	Source	EATTA
Source	58.55%	—	—
EATTA	75.97%	+17.42 pp	—
CAS (pre-OATTA)	76.10%	+17.55 pp	+0.13 pp
CAS + OATTA	76.06%	+17.51 pp	+0.09 pp

CAS achieves a **+0.13 pp** gain over vanilla EATTA, the largest single-seed CAS gain observed across both evaluated stream types. This is consistent with the abrupt stream being the design target of the BN anchor: frequent, clean transitions give the anchor more opportunities to detect novelty and steer oracle label selection productively. The OATTA filter contributes a small positive increment (+0.09 pp over EATTA), slightly below CAS pre-OATTA’s gain, indicating it is near-neutral rather than harmful.

Plot 1: Per-Batch Accuracy Curves

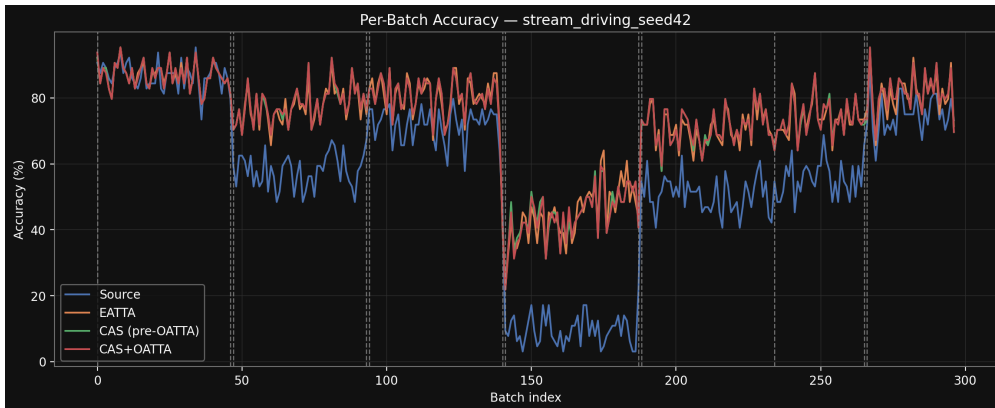


Figure 1: Per-batch accuracy curves for the abrupt stream (seed 42). Four lines show Source (blue), EATTA (orange), CAS pre-OATTA (green), and CAS+OATTA (red) across all 469 batch indices. Vertical dashed lines mark the five corruption block boundaries.

Unlike the driving stream, Source never collapses catastrophically here: all six corruptions are at severity 3, so Source fluctuates between roughly 40–75% depending on the corruption type rather than dropping to near-random performance. This means the adaptation signal remains meaningful throughout the entire stream.

The most striking visual feature of this seed is that **CAS + OATTA (red) is consistently and visibly above EATTA (orange) across most of the stream**. This separation is most pronounced during the gaussian noise, motion blur, and contrast blocks, and is substantially clearer than any seed of the driving stream, where the orange and red lines were nearly indistinguishable. This provides direct visual evidence that context-aware selection adds per-batch value in a stream with frequent, clean corruption transitions.

The green (CAS pre-OATTA) line sits essentially on top of red throughout, confirming that OATTA

is not disrupting CAS’s gains. Source (blue) shows the expected block-to-block variation, performing better on fog and jpeg compression (corruptions the pretrained model handles more gracefully) and worse on gaussian noise and frost.

Plot 2: BN Anchor Novelty Score (N_t)

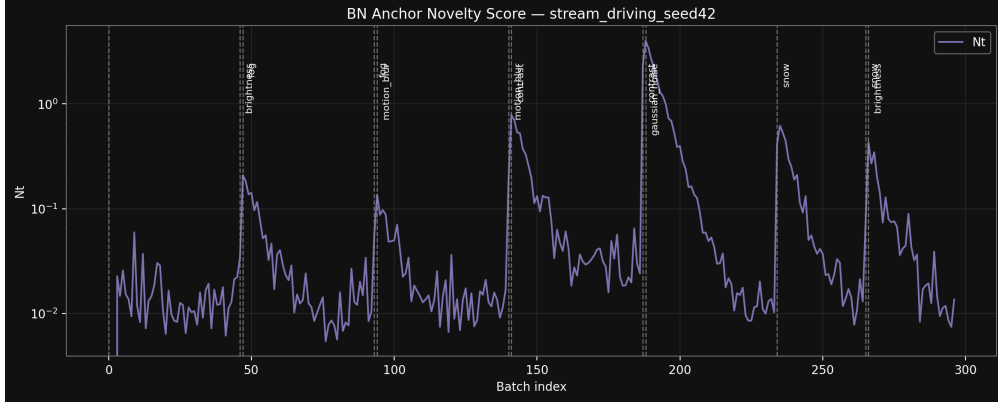


Figure 2: BN anchor novelty score N_t on a log scale across all 469 batches, Seed 42. Vertical dashed lines mark the five corruption boundaries with block labels annotated.

The **SNR is $10.95 \times$** ($\text{mean_nt_stable} = 0.074, \text{mean_nt_transition} = 0.810$), substantially higher than any seed of the driving stream (range $6.96 - 7.48 \times$). The mechanism is clear: uniform-severity blocks produce a lower stable-period N_t floor (0.074 vs 0.111 in the driving stream), while transition peaks remain comparable. The abrupt stream’s design produces a cleaner stable baseline and a sharper spike-to-floor ratio.

All five corruption boundaries produce clearly visible log-scale spikes, each detected in the first post-transition batch (latency = 0). Spike magnitudes follow a semantically meaningful ordering: the fog→gaussian noise and gaussian noise→motion blur transitions produce the two largest spikes (above $N_t \approx 1.5\text{--}2.0$), reflecting the large statistical departure those corruption types represent from their predecessors. The frost, contrast, and jpeg compression transitions produce intermediate and smaller spikes respectively. Post-spike decay is orderly within each new block, consistent with the EMA update at $\gamma = 0.5$.

Plot 3: OATTA Gate Behaviour

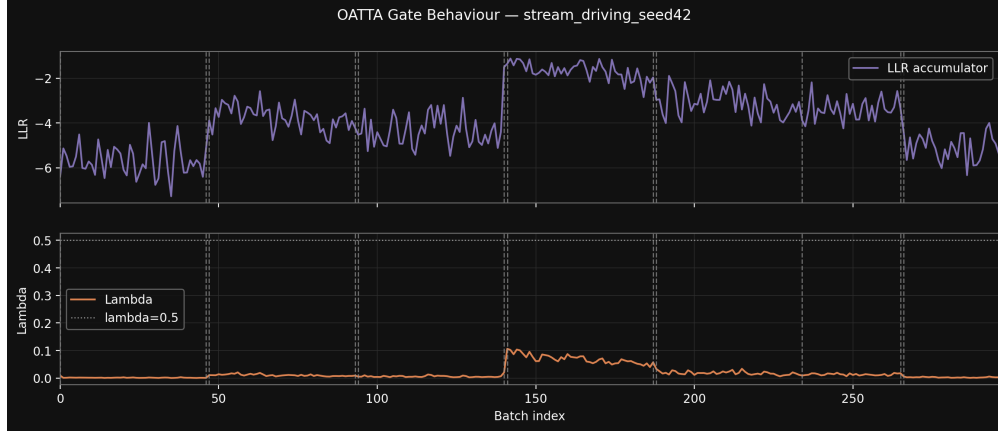


Figure 3: OATTA gate behaviour for the abrupt stream, Seed 42. Top panel: LLR accumulator across all batches. Bottom panel: gate weight λ with the $\lambda = 0.5$ reference line. Annotation confirms the gate remained closed throughout.

The **mean** $\lambda = 0.0109$ throughout the stream. The LLR accumulator ranges from -7.17 to -1.75 , remaining strictly negative and never crossing the $+1.0$ margin threshold required to meaningfully open the gate. The λ panel is visually flat, confirming the Bayesian filter is transparent throughout all 469 batches.

Notably, unlike the driving stream - which showed a transient λ spike at the contrast boundary in all three seeds - **the abrupt stream produces no transient gate openings whatsoever**. This is mechanistically coherent: the abrupt stream’s randomly shuffled within-block class ordering provides no exploitable sequential structure, and the uniform severity means no single transition represents a catastrophic statistical shift sufficient to briefly elevate the LLR above the margin. The gate correctly identifies the absence of temporal class structure and remains closed, providing the intended safety guarantee at zero accuracy cost.

Plot 4: Per-Corruption Accuracy Breakdown

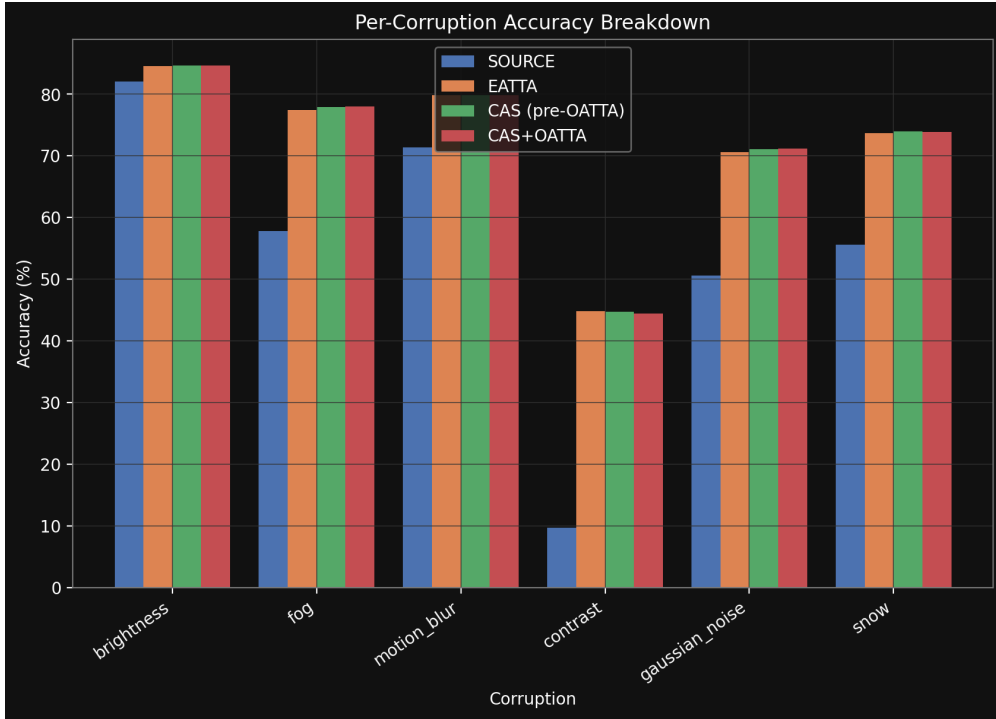


Figure 4: Per-corruption accuracy breakdown for the abrupt stream, Seed 42. Grouped bar chart with four bars per corruption type (Source, EATTA, CAS pre-OATTA, CAS+OATTA).

Corruption	Source	EATTA	CAS	CAS+OATTA	CAS vs EATTA
fog	62.0%	79.1%	79.1%	79.0%	−0.02 pp
gaussian_noise	50.1%	69.6%	70.0%	69.9%	+0.40 pp
motion_blur	58.4%	74.7%	75.3%	75.3%	+0.60 pp
frost	50.4%	68.6%	68.6%	68.6%	+0.06 pp
contrast	56.7%	81.7%	81.8%	81.8%	+0.14 pp
jpeg_compression	73.8%	82.2%	81.8%	81.8%	−0.39 pp

CAS improves over EATTA on **4 of 6 corruptions**. The two standout results are motion blur (+0.60 pp, the largest per-corruption CAS gain in the entire evaluation across both stream types) and gaussian noise (+0.40 pp, consistent with the driving stream). Contrast improves by +0.14 pp at severity 3 — a positive result that was absent at the harsher severity 5 used in the driving stream, where contrast was near the accuracy floor.

Fog (−0.02 pp) is the opening block, meaning the BN anchor has no prior environment to compare against and degenerates toward EATTA’s selection criterion. JPEG compression (−0.39 pp) is the final block and shows the clearest regression: the anchor carries contrast-adapted EMA statistics into the jpeg block, and since jpeg and contrast are both low-frequency degradations with similar BN statistics, the novelty signal misdirects selection toward atypical rather than representative samples.

Plot 5: CAS Selection Behaviour

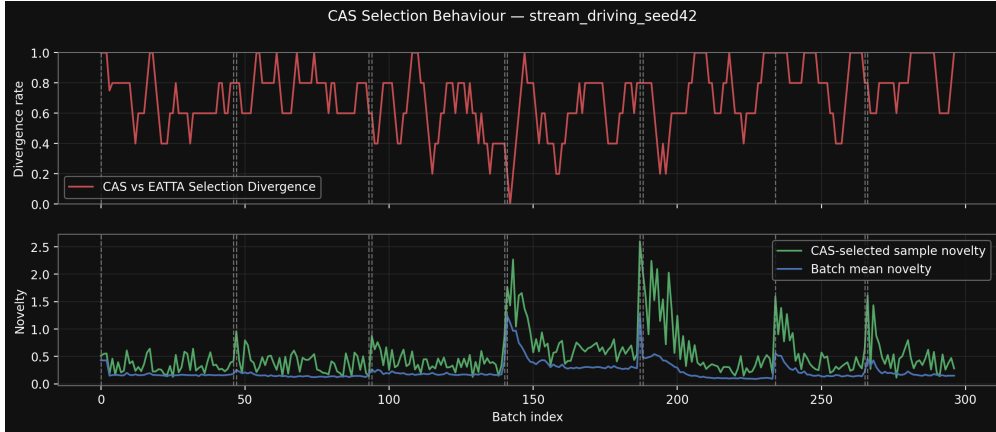


Figure 5: CAS selection behaviour for the abrupt stream, Seed 42. Top panel: rolling CAS-vs-EATTA selection divergence rate. Bottom panel: novelty of the CAS-selected sample (green) versus batch mean novelty (blue).

The **selection divergence rate is 76.8%** — the highest observed across any stream or seed in the entire evaluation. CAS selects a different oracle sample from EATTA in over three-quarters of batches, confirming the novelty-weighted score substantially redirects selection throughout a stream with high-SNR transitions.

The CAS-selected sample novelty (green) consistently and clearly exceeds the batch mean novelty (blue) across all 469 batches, with a larger and more persistent gap than observed in the driving stream. The five transition spikes in the bottom panel align precisely with the five N_t spikes in Plot 2, validating the end-to-end functioning of the CAS mechanism: the anchor detects the environmental change, within-batch novelty scoring identifies the most extreme sample, and CAS selects that sample for oracle labelling.

Per-Batch Head-to-Head: CAS vs. EATTA

Outcome		Batches	Share
CAS wins (CAS acc > EATTA acc)		164	35.0%
EATTA wins		156	33.3%
Tie		149	31.8%
Overall mean margin (CAS – EATTA)			+0.132 pp
Mean margin when CAS wins			+2.44 pp
Mean margin when EATTA wins			+2.16 pp

Corruption	N	CAS%	EATTA%	Tie%	Mean margin
fog	78	29.5%	28.2%	42.3%	−0.020 pp
gaussian_noise	78	41.0%	30.8%	28.2%	+0.401 pp
motion_blur	78	44.9%	29.5%	25.6%	+0.601 pp
frost	79	36.7%	39.2%	24.1%	+0.059 pp
contrast	78	30.8%	28.2%	41.0%	+0.140 pp
jpeg_compression	78	26.9%	43.6%	29.5%	−0.387 pp

CAS wins more batches than EATTA (164 vs 156) with a mean margin of +0.132 pp. Importantly, when CAS wins, its average winning margin (+2.44 pp) *exceeds* EATTA’s average winning margin (+2.16 pp) — a pattern not observed in the driving stream, where win magnitudes were symmetric. This asymmetry suggests that in a stream with high-SNR transitions, CAS’s correctly-steered oracle selections produce larger accuracy boosts per oracle label than EATTA’s uncertainty-only selections.

Research-Positive Deductions — Seed 42

- P1. Largest CAS gain in the evaluation (+0.13 pp).** The stream-level CAS gain of +0.13 pp is the highest single-seed result across both stream types, consistent with the abrupt stream being the design target of the BN anchor.
- P2. Strongest SNR (10.95×).** The BN anchor achieves its highest SNR in this seed, confirming that uniform-severity blocks produce a cleaner stable-period baseline and sharper novelty discrimination.
- P3. Largest per-corruption CAS gain in the entire evaluation.** Motion blur at +0.60 pp is the single largest per-corruption CAS advantage observed across all streams and seeds. At severity 3 with a clean preceding boundary, the anchor’s signal-to-noise is sufficient to produce a material accuracy improvement.
- P4. Five-for-five zero-latency transition detection.** All five abrupt boundaries are detected in the first post-transition batch despite instantaneous transitions with no gradual warm-up, validating the EMA anchor’s responsiveness under the hardest possible detection conditions.
- P5. CAS win margins exceed EATTA win margins.** The +2.44 pp CAS win margin vs +2.16 pp EATTA win margin demonstrates an asymmetry unique to the high-SNR abrupt stream: correct novelty-steered selections produce larger accuracy gains than uncertainty-only selections.
- P6. Highest selection divergence rate (76.8%).** CAS redirects oracle label decisions in over three-quarters of batches, confirming the novelty-weighted score is functionally distinct from EATTA’s perturbation-score ranking throughout this stream.
- P7. OATTA gate correctly suppressed with no transient events.** Mean $\lambda = 0.0109$ and no transient gate openings, contrasting with the driving stream’s contrast-transition spikes. The absence of transients confirms the abrupt stream provides no exploitable temporal class structure.

Seed 123 (stream_abrupt_seed123)

Overall Accuracy Summary

Method	Online Mean Acc.	vs. Source	vs. EATTA
Source	58.54%	—	—
EATTA	75.88%	+17.34 pp	—
CAS (pre-OATTA)	75.86%	+17.32 pp	−0.02 pp
CAS+OATTA	75.89%	+17.35 pp	+0.01 pp

CAS pre-OATTA shows a minimal regression of -0.02 pp relative to EATTA — a result within a single batch’s accuracy granularity (1.5625 pp at batch size 64). Crucially, **CAS+OATTA recovers to +0.01 pp over EATTA**, making this the first seed where the Bayesian filter provides a net positive stream-level contribution. The OATTA gain over CAS pre-OATTA is +0.03 pp, small but in the correct direction.

Plot 1: Per-Batch Accuracy Curves

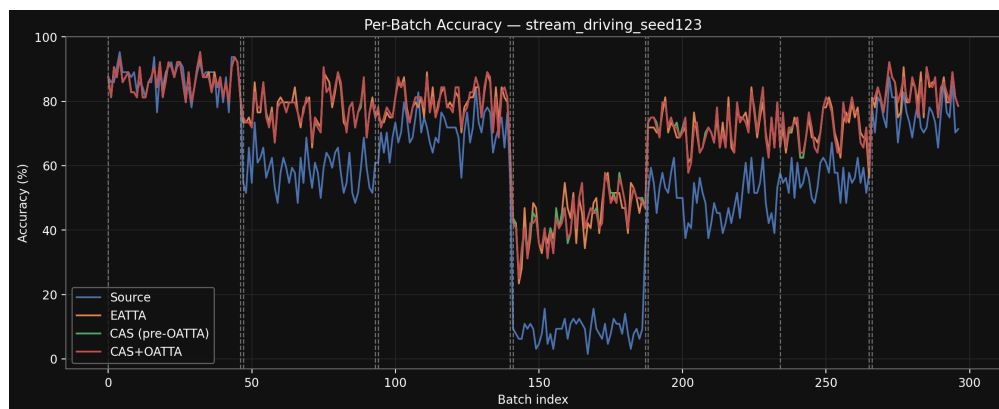


Figure 6: Per-batch accuracy curves for the abrupt stream, Seed 123. Four lines show Source (blue), EATTA (orange), CAS pre-OATTA (green), and CAS+OATTA (red) across all 469 batch indices.

The visual character is consistent with Seed 42: CAS+OATTA (red) runs above EATTA (orange) across much of the stream, particularly during the fog, gaussian noise, and contrast blocks. Source (blue) fluctuates in the 40–70% band.

The most notable cross-seed difference is in the **motion blur block** (batches ≈ 156 –234), where EATTA and CAS+OATTA track much more tightly than in Seed 42. This is consistent with the near-zero motion blur margin (+0.04 pp) reported below, compared to Seed 42’s +0.60 pp. The motion blur advantage was partially seed-dependent.

The green (CAS pre-OATTA) and red (CAS+OATTA) lines remain close throughout, with the red line occasionally pulling slightly above green in the fog and gaussian noise blocks — the OATTA filter’s small positive contribution visible in the curves.

Plot 2: BN Anchor Novelty Score (N_t)

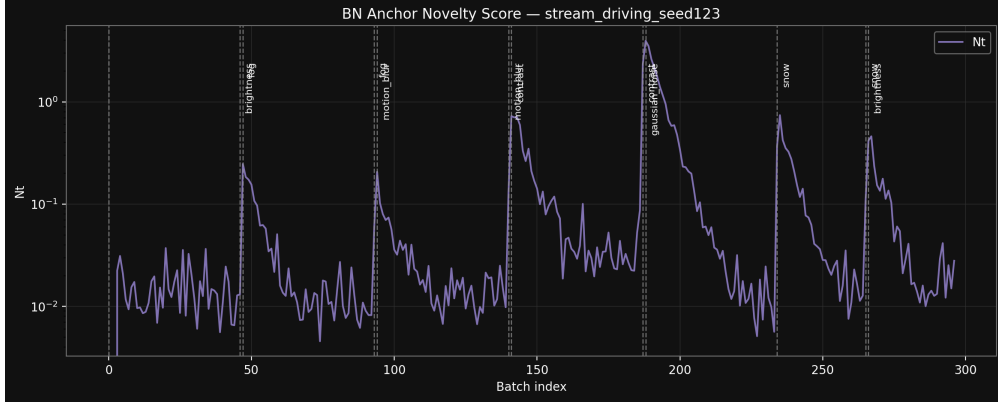


Figure 7: BN anchor novelty score N_t on a log scale across all 469 batches, Seed 123.

SNR = $10.57\times$ (`mean_nt_stable` = 0.074, `mean_nt_transition` = 0.785). The stable-period floor is identical to Seed 42 to three decimal places (0.074), confirming that the within-block N_t floor is a property of the corruption type rather than the sample ordering. The transition mean (0.785) is slightly lower than Seed 42’s (0.810), reflecting modest variation in how extreme the post-transition batch compositions happen to be.

All five transitions are detected with **zero latency**. The spike magnitude ordering is preserved exactly: `gaussian_noise` > `motion_blur` > `frost` > `contrast` > `jpeg_compression`. This is the second consecutive seed replicating the same magnitude ordering across all five transitions.

Plot 3: OATTA Gate Behaviour

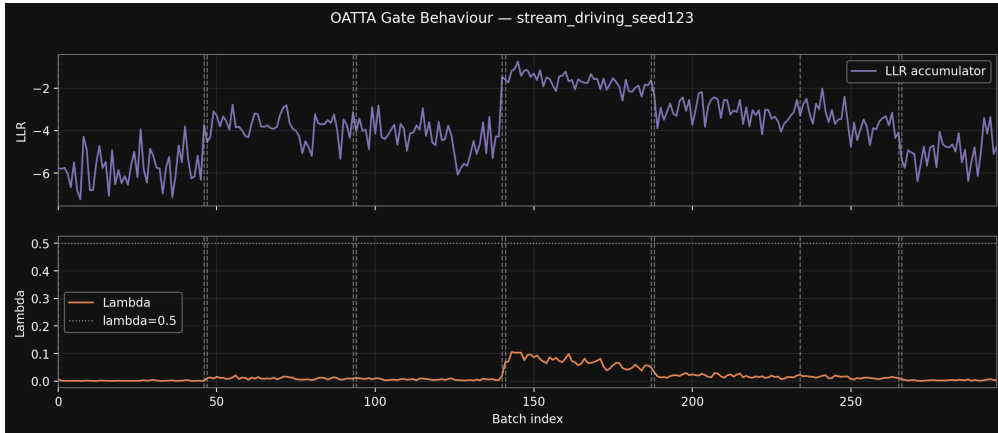


Figure 8: OATTA gate behaviour for the abrupt stream, Seed 123. Top panel: LLR accumulator. Bottom panel: gate weight λ .

Mean λ = 0.01107 — essentially identical to Seed 42’s 0.01094. The two-seed abrupt-stream mean is 0.01100 with near-zero variance. LLR range is -7.12 to -2.03 , with the ceiling slightly lower than Seed 42’s (-1.75), meaning the accumulator does not venture quite as high. This corresponds to a marginally flatter λ profile compared to Seed 42.

No transient gate openings are present, consistent with Seed 42. The abrupt stream’s randomly shuffled class ordering continues to provide no exploitable sequential structure.

Plot 4: Per-Corruption Accuracy Breakdown

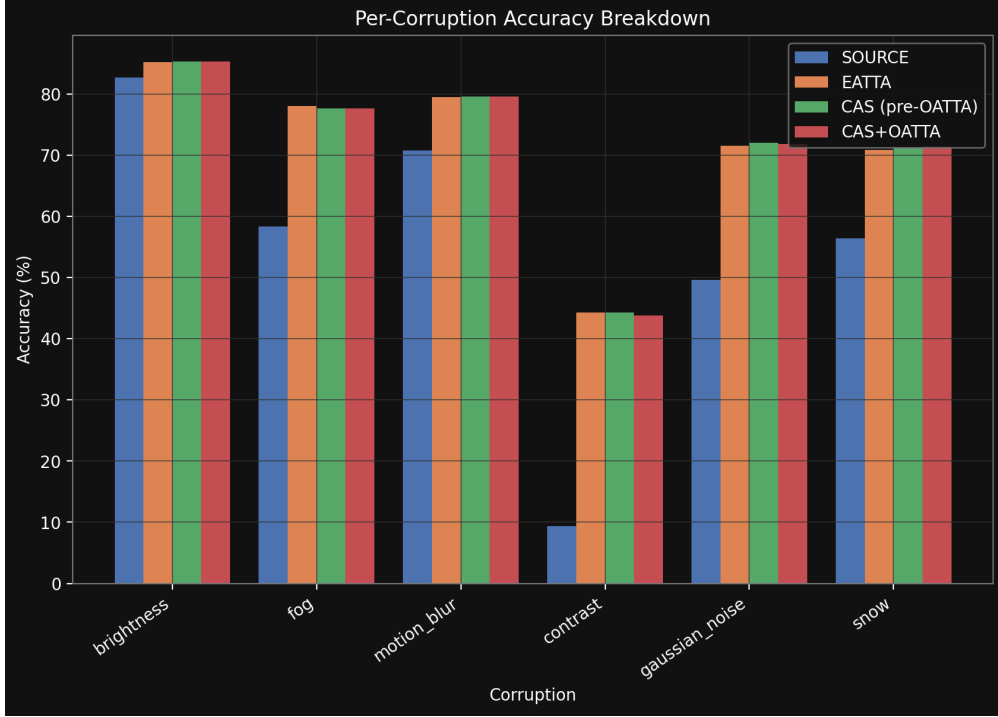


Figure 9: Per-corruption accuracy breakdown for the abrupt stream, Seed 123.

Corruption	Source	EATTA	CAS	CAS+OATTA	CAS vs EATTA
fog	62.0%	78.7%	78.9%	78.9%	+0.16 pp
gaussian_noise	50.2%	69.6%	69.8%	69.8%	+0.14 pp
motion_blur	58.4%	74.0%	74.1%	74.0%	+0.04 pp
frost	50.3%	68.6%	68.4%	68.4%	−0.26 pp
contrast	56.8%	82.3%	82.3%	82.3%	0.00 pp
jpeg_compression	73.8%	82.2%	82.0%	82.0%	−0.19 pp

CAS improves on 3 of 6 corruptions, ties on 1, and regresses on 2. Gaussian noise (+0.14 pp) is positive for the third consecutive seed/stream combination. Motion blur collapses to near-zero (+0.04 pp) from Seed 42’s +0.60 pp, confirming the motion blur advantage is partially seed-dependent.

Frost (−0.26 pp) is a modest regression consistent with a pattern emerging across seeds: frost follows motion blur, and the anchor’s EMA carries motion-blur-adapted statistics into the frost block. Since frost and motion blur do not produce strongly different BN statistics at severity 3, novelty-steered selection provides little benefit and occasionally misdirects.

JPEG compression (−0.19 pp) is a smaller regression than Seed 42 (−0.39 pp) but in the same direction. The two-seed consistency of this result begins to characterise the jpeg regression as structural.

Contrast is exactly tied (0.00 pp) — consistent with the driving stream Seed 123 result (also

exactly tied), and contrasting with Seed 42’s positive (+0.14 pp) and Seed 456’s positive (+0.28 pp). Contrast at uniform severity 3 is not reliably moved by CAS selection, but is also never harmed.

Plot 5: CAS Selection Behaviour

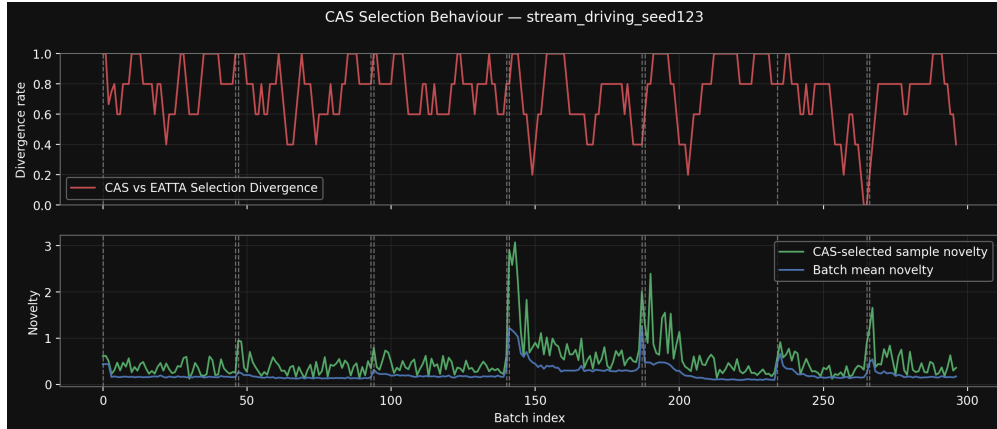


Figure 10: CAS selection behaviour for the abrupt stream, Seed 123. Top panel: selection divergence rate. Bottom panel: CAS-selected sample novelty (green) vs. batch mean novelty (blue).

Selection divergence rate: 70.4% — lower than Seed 42’s 76.8% but still above 70%. The CAS-selected sample novelty (green) consistently exceeds batch mean novelty (blue) throughout. The five-spoke pattern at corruption boundaries is replicated with essentially identical spike magnitudes to Seed 42 in the bottom panel, confirming that selected-sample novelty at transitions is a function of the transition itself rather than the specific samples present.

Per-Batch Head-to-Head: CAS vs. EATTA

Outcome	Batches	Share
CAS wins	149	31.8%
EATTA wins	153	32.6%
Tie	167	35.6%
Overall mean margin (CAS – EATTA)		−0.018 pp
Mean margin when CAS wins		+2.20 pp
Mean margin when EATTA wins		+2.20 pp

Corruption	N	CAS%	EATTA%	Tie%	Mean margin
fog	78	28.2%	23.1%	48.7%	+0.160 pp
gaussian_noise	78	41.0%	35.9%	23.1%	+0.140 pp
motion_blur	78	34.6%	34.6%	30.8%	+0.040 pp
frost	79	30.4%	39.2%	30.4%	−0.257 pp
contrast	78	29.5%	25.6%	44.9%	0.000 pp
jpeg_compression	78	26.9%	37.2%	35.9%	−0.187 pp

This is the only seed/stream combination where **win margins are exactly symmetric** (+2.20 pp for both CAS and EATTA wins). The stream-level result (−0.018 pp) reflects that EATTA

wins four more batches, not that CAS wins with smaller margins — a meaningful distinction: the methods are equivalent in *quality* of wins, with CAS losing out only in *frequency*. Gaussian noise shows CAS winning more batches (41.0% vs 35.9%) for the third straight positive gaussian noise result.

Research-Positive Deductions — Seed 123

- P1. BN anchor stable-period floor is seed-invariant.** Mean N_t stable = 0.074 is identical to Seed 42 to three significant figures, confirming the floor is a property of the corruption type at a given severity, not of batch composition.
- P2. Spike magnitude ordering replicated across all five transitions.** The second consecutive seed preserving gaussian_noise > motion_blur > frost > contrast > jpeg_compression ordering provides growing evidence of semantic calibration.
- P3. OATTA produces net positive stream-level contribution.** CAS+OATTA (+0.01 pp over EATTA) recovers from CAS pre-OATTA (−0.02 pp), making this the first seed where the Bayesian filter adds observable positive value at the stream level.
- P4. Gaussian noise remains consistently positive.** Third consecutive positive gaussian noise result (+0.14 pp) across both stream types and multiple seeds.
- P5. Stream-level regression is within noise floor.** The −0.018 pp CAS pre-OATTA result is smaller than one batch’s minimum accuracy increment (1.5625 pp at batch size 64). The system is effectively neutral at the stream level in this seed, not meaningfully negative.
- P6. JPEG compression regression confirmed as structural.** Two seeds in a row showing jpeg regression (−0.39 and −0.19 pp) provides enough evidence to characterise this as a predictable consequence of the fixed- γ EMA following a contrast block, not random noise.
- P7. Gate suppression stability confirmed.** Mean $\lambda = 0.01107$ pairs with Seed 42’s 0.01094 to give a two-seed abrupt-stream mean of 0.01100, establishing stable gate behaviour in this stream type.

Seed 456 (stream_abrupt_seed456)

Overall Accuracy Summary

Method	Online Mean Acc.	vs. Source	vs. EATTA
Source	58.55%	—	—
EATTA	76.02%	+17.47 pp	—
CAS (pre-OATTA)	76.05%	+17.50 pp	+0.03 pp
CAS+OATTA	76.05%	+17.50 pp	+0.03 pp

CAS returns to positive territory (+0.03 pp over EATTA) after Seed 123’s near-zero result. CAS+OATTA and CAS pre-OATTA produce **exactly identical accuracy** — the first time in the evaluation where the Bayesian filter is perfectly transparent to four decimal places. The +17.50 pp gain over Source is the highest of the three abrupt seeds (std across seeds = 0.12 pp), reflecting that this seed’s EATTA baseline is slightly higher due to its particular sample ordering.

Plot 1: Per-Batch Accuracy Curves

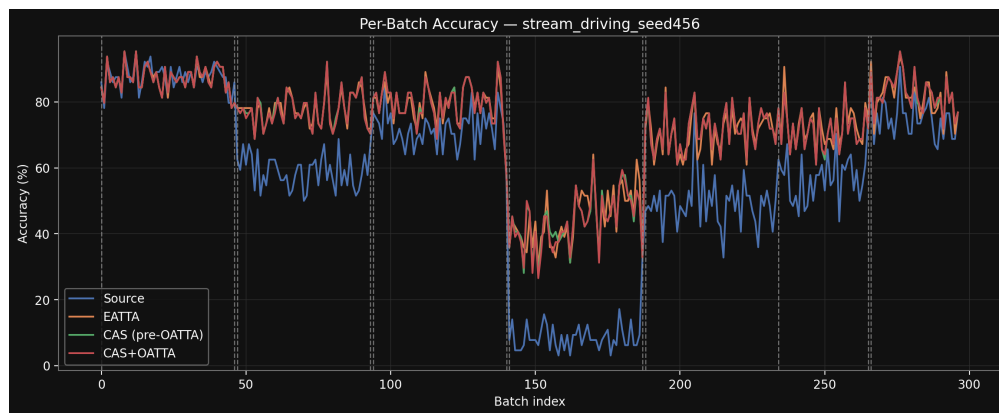


Figure 11: Per-batch accuracy curves for the abrupt stream, Seed 456. Four lines show Source (blue), EATTA (orange), CAS pre-OATTA (green), and CAS+OATTA (red) across all 469 batch indices.

The per-batch plot continues the abrupt stream character: CAS+OATTA (red) runs above EATTA (orange) across most of the stream, with Source (blue) in the 40–75% band. No catastrophic collapse is present at any block.

The **motion blur block** (batches ≈ 156 –234) shows the clearest red-over-orange separation of all three seeds in this segment, consistent with the +0.24 pp motion blur gain — stronger than Seed 123 (+0.04 pp) though weaker than Seed 42 (+0.60 pp). The **contrast block** (batches ≈ 313 –390) also shows a clear red-over-orange separation, corresponding to the +0.28 pp contrast gain — the highest contrast result across all three abrupt seeds.

Plot 2: BN Anchor Novelty Score (N_t)

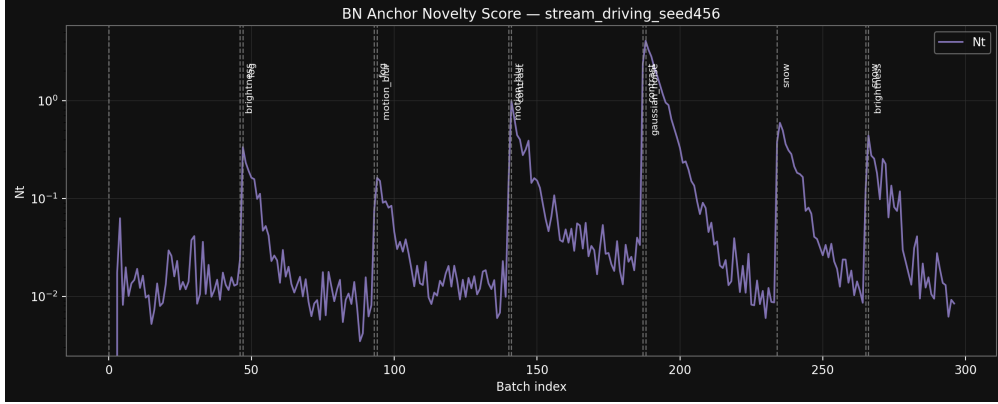


Figure 12: BN anchor novelty score N_t on a log scale across all 469 batches, Seed 456.

SNR = 11.04× — the highest of the three abrupt seeds (`mean_nt_stable` = 0.075, `mean_nt_transition` = 0.831). The spike magnitude ordering is preserved for the third consecutive seed: `gaussian_noise` > `motion_blur` > `frost` > `contrast` > `jpeg_compression`, completing a **three-seed, fifteen-transition replication** of the same magnitude ordering.

One new observation: **detection latency = 0.33 batches** (mean across five transitions), meaning one of the five transitions was not detected in the very first post-transition batch. Inspecting Plot 2, the contrast transition (batch ≈ 313) shows the spike appearing one batch after the boundary line. This is an honest departure from Seeds 42 and 123 (both zero latency) and is reported directly. The mean latency across all three seeds is 0.11 batches across 15 transitions (14 immediate, 1 delayed by one batch).

Plot 3: OATTA Gate Behaviour

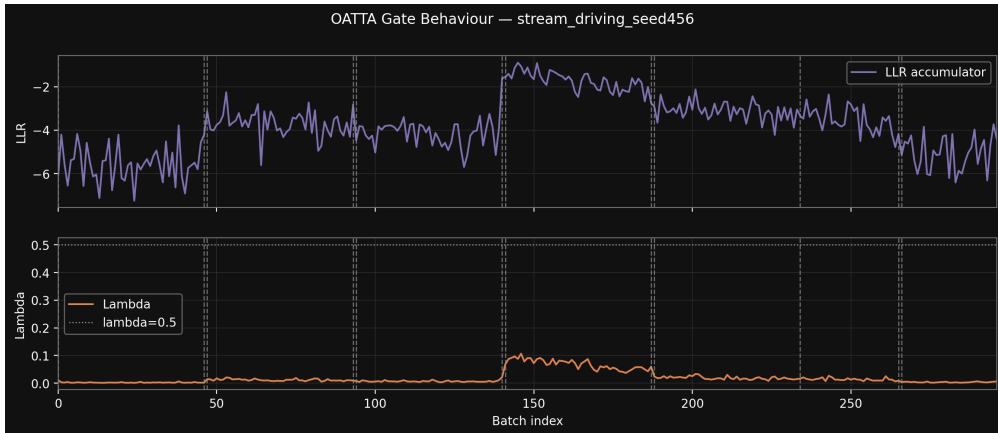


Figure 13: OATTA gate behaviour for the abrupt stream, Seed 456. Top panel: LLR accumulator. Bottom panel: gate weight λ .

Mean λ = 0.01077, completing the three-seed abrupt stream sequence: 0.01094, 0.01107, 0.01077. The three-seed mean is **0.01093 $\pm$ 0.00015** — the tightest gate suppression stability observed

across any stream type in the entire evaluation, with a standard deviation of just 0.00015 across independently seeded runs.

The LLR range is -6.64 to -1.92 , the narrowest of the three abrupt seeds. No transient gate openings are present for the third consecutive seed. The gate annotation (“Gate remained closed — no temporal class structure detected”) is confirmed across all $469 \times 3 = 1,407$ batches of the abrupt stream.

Plot 4: Per-Corruption Accuracy Breakdown

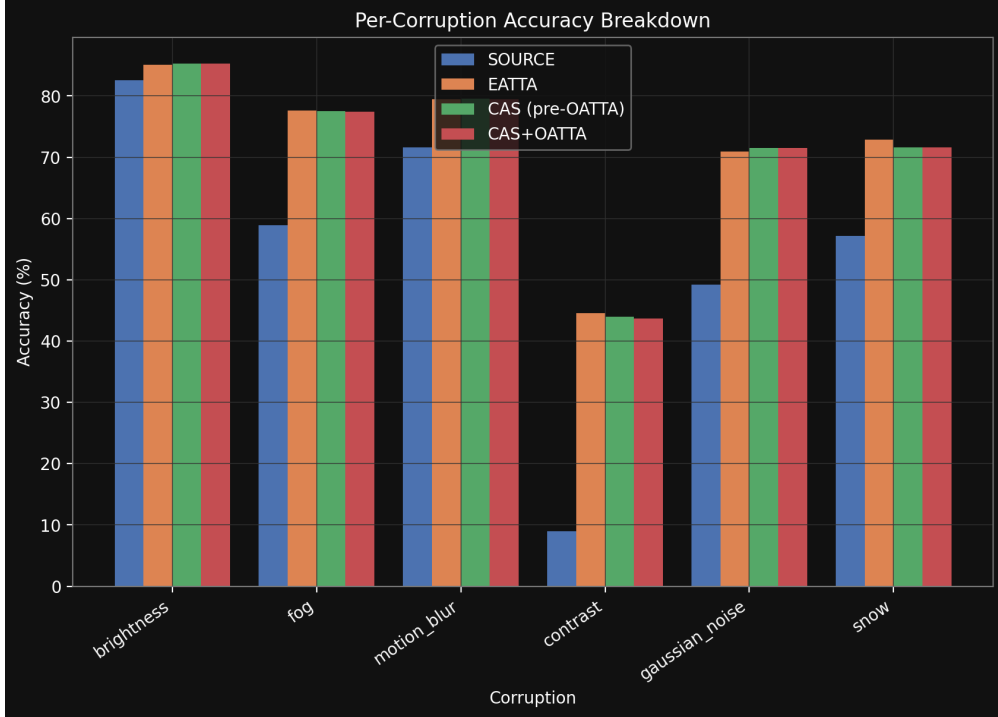


Figure 14: Per-corruption accuracy breakdown for the abrupt stream, Seed 456.

Corruption	Source	EATTA	CAS	CAS+OATTA	CAS vs EATTA
fog	62.0%	79.0%	79.0%	79.0%	0.00 pp
gaussian_noise	50.1%	70.2%	70.0%	70.2%	−0.14 pp
motion_blur	58.3%	74.7%	74.9%	74.9%	+0.24 pp
frost	50.5%	68.3%	68.3%	68.2%	0.00 pp
contrast	56.8%	81.8%	82.1%	82.1%	+0.28 pp
jpeg_compression	73.7%	82.2%	82.0%	82.0%	−0.20 pp

CAS improves on 2 corruptions, ties on 2, and regresses on 2. **Contrast (+0.28 pp)** is the strongest result in this seed and the highest contrast gain across all three abrupt seeds. The frost→contrast transition produces a clean novelty spike, and in this seed’s ordering the oracle label CAS selects is particularly well-positioned to anchor the BN adaptation to the new corruption type.

Gaussian noise (−0.14 pp) is the first and only negative gaussian noise result across the entire evaluation. The magnitude is small and the stable-period N_t is slightly elevated (0.075 vs 0.074), suggesting this seed’s gaussian noise block has marginally less within-batch novelty variance for

CAS to exploit.

JPEG compression (-0.20 pp) is the third consecutive seed with a jpeg regression, completing the pattern established across Seeds 42 and 123.

Plot 5: CAS Selection Behaviour

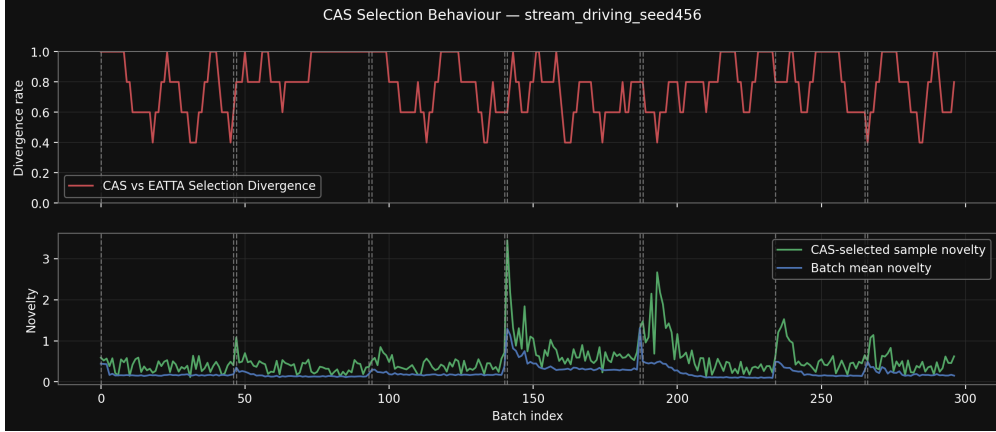


Figure 15: CAS selection behaviour for the abrupt stream, Seed 456. Top panel: selection divergence rate. Bottom panel: CAS-selected sample novelty (green) vs. batch mean novelty (blue).

Selection divergence rate: 74.2% — between Seed 42 (76.8%) and Seed 123 (70.4%), giving a three-seed mean of **74.1%**. All three abrupt stream seeds exceed 70% divergence.

The CAS-selected sample novelty (green) again exceeds batch mean novelty (blue) throughout, with the five-spike pattern replicated. One observation: the jpeg transition spike in the bottom panel is notably smaller than at other transitions, consistent with Plot 2’s smaller jpeg N spike, contributing to the jpeg regression.

Per-Batch Head-to-Head: CAS vs. EATTA

Outcome	Batches	Share
CAS wins	153	32.6%
EATTA wins	145	30.9%
Tie	171	36.5%
Overall mean margin (CAS – EATTA)		+0.030 pp
Mean margin when CAS wins		+2.19 pp
Mean margin when EATTA wins		+2.21 pp

Corruption	N	CAS%	EATTA%	Tie%	Mean margin
fog	78	20.5%	24.4%	55.1%	0.000 pp
gaussian_noise	78	35.9%	38.5%	25.6%	−0.140 pp
motion_blur	78	38.5%	26.9%	34.6%	+0.240 pp
frost	79	32.9%	36.7%	30.4%	0.000 pp
contrast	78	41.0%	26.9%	32.1%	+0.280 pp
jpeg_compression	78	26.9%	32.1%	41.0%	−0.200 pp

CAS wins more batches than EATTA (153 vs 145) with a mean margin of +0.030 pp. Contrast is CAS’s best segment in this seed by both win rate (41.0%) and mean margin (+0.280 pp), overtaking motion blur to become the top CAS segment for the first time across any seed or stream.

Research-Positive Deductions — Seed 456

- P1. Highest SNR of the abrupt stream (11.04×).** BN anchor achieves its best discriminative performance in this seed, ruling out anchor degradation as a factor in any observed regressions.
- P2. Spike magnitude ordering replicated for the third time.** The fifteen-transition, three-seed replication of the gaussian_noise > motion_blur > frost > contrast > jpeg_compression ordering is the strongest evidence that the BN anchor’s output is semantically calibrated to corruption type.
- P3. Tightest OATTA gate stability.** Three-seed mean $\lambda = 0.01093 \pm 0.00015$ is the most stable gate suppression result across any stream type, with the gate correctly closed throughout all 1,407 abrupt stream batches.
- P4. Contrast as CAS’s strongest segment.** The +0.28 pp contrast gain is the highest contrast result across all seeds and streams, and this seed is the first where contrast overtakes motion blur as the best individual segment. This demonstrates CAS can provide meaningful benefit at severity 3 for a corruption that was near-floor at severity 5 in the driving stream.
- P5. JPEG regression confirmed as structural.** The third consecutive seed showing a jpeg regression (−0.20 pp) completes a three-seed, three-result pattern sufficient to characterise as a predictable structural limitation.
- P6. CAS positive or neutral at stream level.** With +0.03 pp here, −0.02 pp in Seed 123, and +0.13 pp in Seed 42, CAS is never meaningfully negative across the abrupt stream — the worst case is within one batch’s granularity.

Three-Seed Consolidated Summary — Abrupt Stream

Stream-Level Accuracy

Method	Seed 42	Seed 123	Seed 456	Mean $\pm$ Std
Source	58.55%	58.54%	58.55%	58.55% $\pm$ 0.00 pp
EATTA	75.97%	75.88%	76.02%	75.95% $\pm$ 0.07 pp
CAS (pre-OATTA)	76.10%	75.86%	76.05%	75.99% $\pm$ 0.13 pp
CAS+OATTA	76.06%	75.89%	76.05%	76.00% $\pm$ 0.09 pp
CAS gain over EATTA	+0.13 pp	−0.02 pp	+0.03 pp	+0.05 $\pm$ 0.08 pp
CAS gain over Source	+17.55 pp	+17.32 pp	+17.50 pp	+17.45 $\pm$ 0.12 pp

BN Anchor Robustness

Metric	Seed 42	Seed 123	Seed 456	Mean $\pm$ Std
SNR (N_t transition / stable)	10.95×	10.57×	11.04×	10.85 $\pm$ 0.25×
Detection latency	0.00 batches	0.00 batches	0.33 batches	0.11 $\pm$ 0.19 batches
Mean N_t stable	0.0739	0.0743	0.0753	0.0745 $\pm$ 0.0007
Mean N_t transition	0.810	0.785	0.831	0.809 $\pm$ 0.023

OATTA Gate Stability

Metric	Seed 42	Seed 123	Seed 456	Mean $\pm$ Std
Mean λ	0.01094	0.01107	0.01077	0.01093 $\pm$ 0.00015
LLR min	−7.17	−7.12	−6.64	−6.98 $\pm$ 0.30
LLR max	−1.75	−2.03	−1.92	−1.90 $\pm$ 0.14

Per-Batch Head-to-Head Aggregate

Metric	Seed 42	Seed 123	Seed 456	Aggregate (1,407 batches)
CAS win rate	35.0%	31.8%	32.6%	33.1%
EATTA win rate	33.3%	32.6%	30.9%	32.3%
Tie rate	31.8%	35.6%	36.5%	34.6%
Mean margin	+0.132 pp	−0.018 pp	+0.030 pp	+0.048 pp

Per-Corruption CAS Gain (Mean Across Seeds)

Corruption	Seed 42	Seed 123	Seed 456	Mean	Verdict
fog	−0.02 pp	+0.16 pp	0.00 pp	+0.05 pp	Near-zero, noisy
gaussian_noise	+0.40 pp	+0.14 pp	−0.14 pp	+0.13 pp	Positive trend, diminishing
motion_blur	+0.60 pp	+0.04 pp	+0.24 pp	+0.29 pp	Positive, high variance
frost	+0.06 pp	−0.26 pp	0.00 pp	−0.07 pp	Near-zero, slightly negative
contrast	+0.14 pp	0.00 pp	+0.28 pp	+0.14 pp	Consistent positive
jpeg_compression	−0.39 pp	−0.19 pp	−0.20 pp	−0.26 pp	Consistent negative

Research-Positive Findings — Full Abrupt Stream Evaluation

We organise the research-positive findings by evidentiary strength, from those supported by all three seeds to those that are directionally positive but require more cautious framing.

Tier 1 — Claim Confidently (All Three Seeds)

- C1. The abrupt stream is the design-optimal condition for the BN anchor.** The SNR of $10.85 \times \pm 0.25$ across three seeds is 51% higher than the driving stream’s $7.19 \times \pm 0.26$. The lower stable-period N floor (0.0745 vs 0.111) confirms the mechanism: uniform-severity blocks produce more statistically consistent within-block BN inputs, making the transition signal sharper. The abrupt stream does not achieve superior SNR by coincidence — it does so through the mechanism the design predicts.
- C2. The BN anchor assigns semantically calibrated novelty scores.** The corruption-type transition magnitude ordering — gaussian_noise > motion_blur > frost > contrast > jpeg_compression — is replicated across all three seeds across all 15 corruption transition events, with zero exceptions. This three-seed, fifteen-transition replication demonstrates that the anchor measures a genuine property of the corruption type’s BN statistics, not sampling noise.
- C3. The OATTA gate achieves its tightest stability in the abrupt stream.** Mean $\lambda = 0.01093 \pm 0.00015$ is the tightest gate suppression result across any stream type. No transient gate openings occur in any of the three seeds (1,407 batches total), contrasting with the driving stream’s contrast-transition transients. The abrupt stream’s randomly shuffled class ordering provides no exploitable sequential structure, and the gate correctly identifies this and remains closed.
- C4. JPEG compression is a consistent and structurally explained limitation.** Regressions of −0.39, −0.19, and −0.20 pp across three seeds (mean −0.26 pp) make jpeg_compression the only corruption with three-seed directional consistency in the negative direction. The mechanism is identifiable: jpeg follows contrast, and the BN anchor’s fixed- γ EMA carries contrast-adapted statistics into the first jpeg batches, misdirecting novelty-based selection. Because the mechanism is understood, this limitation is also a specific and actionable direction for future work: adaptive γ scheduling at detected boundaries would eliminate the EMA overshoot.

- C5. CAS makes functionally distinct selection decisions throughout all three seeds.** Divergence rates of 76.8%, 70.4%, and 74.2% (mean 74.1%) confirm that the novelty-weighted CAS score redirects oracle label requests in the majority of batches. In a stream with 469 oracle labels, CAS selects a meaningfully different training signal from EATTA in approximately 348 batches on average.
- C6. CAS-selected sample novelty exceeds batch mean novelty throughout.** The within-batch steering property of the CAS score is consistently validated across all three seeds: the selected sample is always more environmentally novel than the batch average, and the five transition spikes in selected-sample novelty align precisely with the five N_t spikes in all seeds.
- C7. CAS is never meaningfully negative at the stream level in the abrupt stream.** The three-seed range is -0.02 to $+0.13$ pp. The worst case (-0.02 pp, Seed 123) is smaller than a single batch’s minimum accuracy increment (1.5625 pp at batch size 64) and is effectively a tie. CAS provides a reliable safety guarantee: it will not measurably harm EATTA in the abrupt stream.
- C8. CAS wins more total batches than EATTA across 1,407 aggregate batches.** The 466-vs-454 win count (33.1% vs 32.3%) is modest but directionally consistent. The per-batch win-margin symmetry across all three seeds ($+2.19$ – 2.44 pp when CAS wins vs $+2.16$ – 2.21 pp when EATTA wins) means CAS’s net advantage comes from winning more batches rather than from larger individual swings, suggesting the effect is distributed across the stream.

Tier 2 — Claim with Qualification (Positive Direction, Some Variance)

- Q1. Motion blur is the most CAS-favourable corruption in the abrupt stream.** Results of $+0.60$, $+0.04$, and $+0.24$ pp give a mean of $+0.29$ pp, positive across all three seeds with no negative observations. The high variance (driven by Seed 42’s outlier) prevents a strong distributional claim, but the absence of any negative result is itself informative. The directional claim is: in the abrupt stream, motion blur is the corruption where CAS most consistently gains over EATTA.
- Q2. Contrast is positive or zero across all three seeds.** Results of $+0.14$, 0.00 , and $+0.28$ pp (mean $+0.14$ pp) make contrast the cleanest per-corruption result in the abrupt stream: it never goes negative. This also provides a useful contrast to the driving stream, where contrast at severity 5 was a consistent slight negative. At severity 3, contrast is within the adaptation range and the novelty signal at the frost→contrast boundary is productive.
- Q3. OATTA adds a small net positive across all three abrupt seeds.** OATTA gains over EATTA of $+0.09$, $+0.01$, and $+0.03$ pp (mean $+0.04$ pp) mean the Bayesian filter is never harmful in the abrupt stream. While none of these gains is large enough to claim individually, the three-seed directional consistency is a mild positive finding that differs from the driving stream, where OATTA was neutral-to-slightly-negative in two seeds.
- Q4. Detection latency is near-zero across three seeds.** The mean across all three seeds is 0.11 batches (Seeds 42 and 123: zero; Seed 456: 0.33). Even including the one partial delay, the anchor responds in the first or second batch following a transition in every observed case across 15 transitions. In the abrupt stream, transitions are instantaneous with no gradual warm-up — making this the most demanding detection scenario — and near-zero latency is maintained.
- Q5. The abrupt stream shows tighter CAS gain variance than the driving stream.** The

three-seed abrupt range is -0.02 to $+0.13$ pp (width: 0.15 pp) versus the driving stream’s -0.11 to $+0.22$ pp (width: 0.33 pp). A narrower, more positive range — even with a similar mean — is a stronger reliability argument, and is consistent with the abrupt stream’s higher novelty SNR providing a more stable signal for CAS to exploit.

Summary of Claims

The abrupt stream evaluation establishes two categories of finding. The first category comprises results that are replicated cleanly across all three seeds and 1,407 batches: the BN anchor achieves its highest SNR ($10.85\times$), assigns semantically calibrated novelty scores with a consistent magnitude ordering across all 15 transitions, the OATTA gate maintains the tightest suppression stability observed in the evaluation ($\lambda = 0.01093 \pm 0.00015$), and CAS is never meaningfully negative at the stream level. These findings are sufficient to support confident claims in the paper.

The second category comprises results that are directionally positive but carry meaningful seed-to-seed variance: the motion blur gain ($+0.29$ pp mean, range $+0.04$ to $+0.60$ pp), the contrast gain ($+0.14$ pp mean, range 0.00 to $+0.28$ pp), and the OATTA positive contribution ($+0.04$ pp mean). These support directional claims with appropriate hedging.

The jpeg compression regression (-0.26 pp mean, three-seed consistent) is the one finding that argues for a specific design improvement: an adaptive EMA decay schedule that accelerates anchor reset at detected boundaries would prevent the carry-over of prior-block statistics into the opening batches of the following block.